RAJALAKSHMI ENGINEERING COLLEGE

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V2

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|---|---|
| **Started on** | Wednesday, 16 September 2026, 8:51 AM |
| **State** | Finished |
| **Completed on** | Wednesday, 16 September 2026, 8:57 AM |
| **Time taken** | 6 mins 2 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

Given an array of N integer, we have to maximize the sum of arr[i] * i, where i is the index of the element (i = 0, 1, 2, ..., N).Write an algorithm based on Greedy technique with a Complexity O(nlogn).

Input Format:

First line specifies the number of elements-n

The next n lines contain the array elements.

Output Format:

Maximum Array Sum to be printed.

Sample Input:

5

2 5 3 4 0

Sample output:

40

**Answer:** (penalty regime: 0 %)

```c
#include<stdio.h>
int main(){
    int n;
    scanf("%d",&n);
    int a[n];
    for(int i=0;i<n;i++){
        scanf("%d",&a[i]);
    }
    for(int i=0;i<n-1;i++){
        for(int j=0;j<n-i-1;j++){
            if(a[j]>a[j+1]){
                int temp=a[j];
                a[j]=a[j+1];
                a[j+1]=temp;
            }
        }
    }
    long long sum=0;
    for(int i=0;i<n;i++){
        sum+=(long long)a[i]*i;
    }
    printf("%lld",sum);
    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 5<br>2<br>5<br>3<br>4<br>0 | 40 | 40 | ✔ |

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 10<br>2<br>2<br>2<br>4<br>4<br>3<br>3<br>5<br>5<br>5 | 191 | 191 | ✔ |
| ✔ | 2<br>45<br>3 | 45 | 45 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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